5.4.4 Testing of Sewers

The sewer line laid should be tested before it is put in to use. The tests to be done are as follows

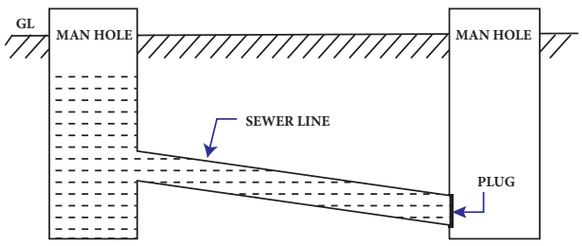
a. Water tightness test
b. Obstruction test
c. Straightness test

a. Water tightness test

After constructing the sewer, some period of time should be given for the joints to become hard and strong.

The water tightness of the joints is carried out between two manholes whether it is capable of withstanding 1.5m of pressure head by the following procedure

- The manhole at lower end of the sewer line is provided with a plug by using rubber bag with canvas cover and it is tightly closed.
- Water is filled in the manhole at the upper end of the sewer line.
- The depth of water in the upper end manhole is maintained at 1.5m.
- If the water level gets reduced, it indicates leakage and the corresponding pipe is replaced.



TESTING OF SEWER LINE

d. Obstruction test

In this test a smooth ball less than the diameter of the sewer pipe is inserted at the high end of the sewer (or) drain. In the absence of obstruction, the ball shall roll down in the invert of the pipe and emerge at the lower end.

c. Straightness test

In this test a mirror is placed at one end and a lamp is placed at the other end. If the pipeline is straight, the full section of the sewer is observed in the mirror, otherwise this would be apparent. The mirror will also indicate any obstruction in the sewer line.

5.4.5 Ventilation of sewers

The ventilation of sewer is essential for the following reasons

1. To avoid nuisance causing unpleasant odours.
2. To avoid the accumulation of explosive and poisonous gases and vapours.
3. To avoid airlocks in the sewers.
4. To relieve air pressure above (or) below atmospheric pressure caused by sudden rise or fall of sewage.

Methods of Ventilation

1. Proper design of sewers
2. Proper construction of sewers
3. Providing manholes with gratings
4. Providing ventilating columns (or) shafts
5. Providing unobstructed outlets

5.4.6 Cleaning of Sewers

Once the sewers laid and buried, they are liable to corrosion, deterioration, erosion, etc. Therefore the sewers should be properly cleaned and maintained in good working condition.

For the following reasons, the sewers are to be cleaned:

1. Breakage of sewers
2. Clogging of sewers
3. Odours in the sewers

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